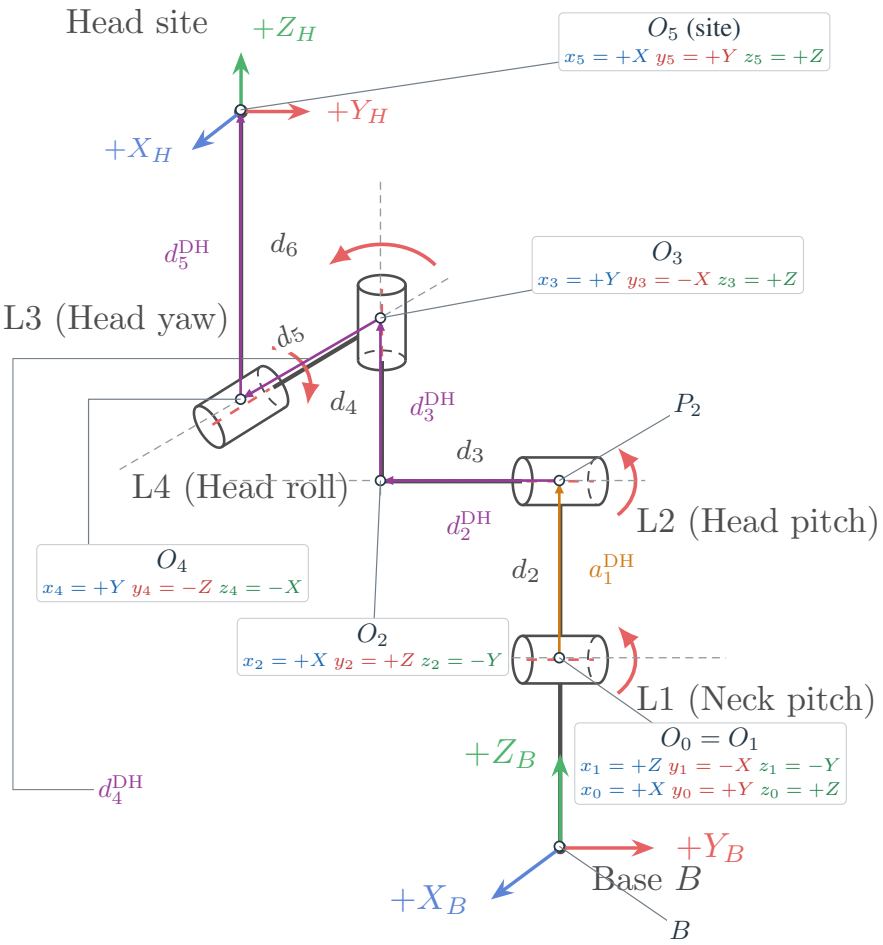
Head and neck: origins, D-H distances and axes

Direct annotations on the original assignment drawing. Zero pose; distances in mm.



Orange: a_{i-1}^{DH} Purple: d_i^{DH}

Original black d labels are handout dimensions. Coloured superscript DH labels are table parameters.

Read each row as a two-step path

$$O_{i-1} \xrightarrow[+x_{i-1}]{a_{i-1}} P_i \xrightarrow[+z_i]{d_i} O_i$$

A negative value means travel opposite to that positive axis.

i	a_{i-1}	d_i	axis for a	axis for d
1	0	0	$+X$	$-Y$
2	66	19.05	$+Z$	$-Y$
3	0	37.85	$+X$	$+Z$
4	0	-5	$+Y$	$-X$
5	0	42.45	$+Y$	$+Z$

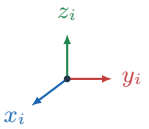
$a_0 = a_2 = a_3 = a_4 = 0$; $d_1 = 0$.
$P_1 = O_0 = O_1$, $P_3 = O_2$, $P_4 = O_3$, $P_5 = O_4$.

Full frame axes, expressed in base B

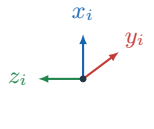
frame	$+x_i$	$+y_i$	$+z_i$
0	$+X$	$+Y$	$+Z$
1	$+Z$	$-X$	$-Y$
2	$+X$	$+Z$	$-Y$
3	$+Y$	$-X$	$+Z$
4	$+Y$	$-Z$	$-X$
5	$+X$	$+Y$	$+Z$

Positive-axis triads
Translated from their O_i

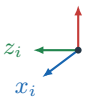
$i = 0, 5$



$i = 1$



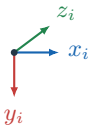
$i = 2$



$i = 3$



$i = 4$



$O_0 = O_1$ at the neck axis; their orientations differ. P_2 is the pitch-axis point vertically above O_1. O_2 is shifted along the head-pitch axis. O_3 and O_4 lie on extended yaw and roll axes. The base-to-neck transform ${}^{B}T_0$ is separate. The handout's $d_1 = 92.021$ mm is not the D-H $d_1 = 0$.

Original: Assignment 1, Fig. 4. Schematic, not to scale; perspective is illustrative. Exact signs and directions are given in the tables. Markers are selected D-H origins, not CAD joint centres.